A SPIN–SCHLÄFLI OBSTRUCTION TO ± 2 COSMETIC SURGERY

Sources: `notes/R8/hyp.md` and `notes/R8/hyp-code/`. K is a hyperbolic knot; by Ren we may assume this, and by Ravelomanana $Y_{\pm 2} = S^3_{\pm 2}(K)$ are then hyperbolic. $\mathrm{cs} \in \mathbb{R}/\mathbb{Z}$ denotes the real part of complex Chern–Simons in $SU(2)$ units. SnapPy's `chern_simons` computes it only modulo $\frac{1}{2}$.

1. Two exact facts

Lemma 1 (spin swap)**.** *$Y_{\pm 2}$ has two spin structures: s_0, which extends over the trace $X_{\pm 2}$, and s_1, which does not. If $\mathrm{Arf}(K) = 0$, then every orientation-preserving homeomorphism $Y_{-2} \to Y_2$ sends s_1 to s_0.*

Proof. By Kirby–Melvin, $\mu(Y_{\pm 2}, s_0) = \pm 1$ and $\mu(Y_{\pm 2}, s_1) = \mp 1 + 8\,\mathrm{Arf}(K)$ in $\mathbb{Z}/16$. A homeomorphism preserves μ, which forces the swap. □

Lemma 2 (half period)**.** *Let $\tilde{\rho}$ be an $SL_2(\mathbb{C})$ representation of $\pi_1(Y_{\pm 2})$, and let ε be the nontrivial character. Then $\mathrm{cs}(\tilde{\rho}\varepsilon) = \mathrm{cs}(\tilde{\rho}) + \frac{1}{2}$.*

Proof. Twisting by ε tensors the flat bundle with the flat line bundle L_ε. For Cheeger–Chern–Simons classes, $\hat{c}_2(E \otimes L) = \hat{c}_2(E) + \hat{c}_1(L) * \hat{c}_1(L)$. Here $\hat{c}_1(L_\varepsilon) * \hat{c}_1(L_\varepsilon)$ is the self-linking of the generator of $H_1 = \mathbb{Z}/2$, namely $\pm\frac{1}{2}$, equivalently $\langle \varepsilon^3, [Y] \rangle / 2$ with $\varepsilon^3 \neq 0$. Checks: on $\mathbb{RP}^3$ the central representation has $\mathrm{cs} = \frac{1}{2}$; on $L(4,1)$, $\varepsilon^3 = 0$. □

Consequence. On a hyperbolic manifold, spin structures are exactly the SL_2 lifts of the holonomy, so $\mathrm{cs}(Y, s)$ is defined modulo 1. Lemma 2 gives $\mathrm{cs}(Y_{\pm 2}, s_1) = \mathrm{cs}(Y_{\pm 2}, s_0) + \frac{1}{2}$. A cosmetic homeomorphism is an isometry (Mostow) and carries s_1 to s_0 (Lemma 1). Hence

$$\mathrm{cs}(Y_2, s_0) - \mathrm{cs}(Y_{-2}, s_0) \equiv \tfrac{1}{2} \pmod 1.$$

The PSL_2 invariants alone only give a value in $\{0, \frac{1}{2}\}$. The spin structure supplies exactly the one missing bit, and it is the bit that a coincidence of PSL_2 data would leave open.

2. Computing the left side from the knot

Lemma 3. *Take the lift of the complete structure on E_K induced by the spin structure of S^3; it has $\mathrm{tr}\,\mu = \mathrm{tr}\,\lambda = -2$. Continue it along the cone deformation for slope $(N, 1)$ with N even. The trace of the filling slope is $-2\cos(\alpha/2)$, which reaches $+2$ at $\alpha = 2\pi$. So the lift descends to Y_N, and the spin structure it determines is s_0.*

Proposition 4 (complex Schläfli)**.** *Write $\tau_N(\alpha) = \mathrm{Im} \log \mathrm{hol}(\mu)$ along the cone path, with $\tau_N(0) = 0$. Then*

$$\mathrm{cs}(Y_N, s_0) \equiv \mathrm{cs}(E_K) + c_N + \tfrac{1}{4\pi^2} \int_0^{2\pi} \tau_N \, d\alpha \pmod 1,$$

where $c_N \equiv N/4 \pmod 1$ is the solid-torus constant. In particular $c_2 = c_{-2}$.

The single point needing an independent check. The dual curve must be the same, namely μ, for both slopes. Replacing it by $\mu \cdot m_{\mathrm{fill}}$ changes the integral by exactly $\frac{1}{2}$, which is precisely the bit at stake. The constants were cross-checked against lens spaces, where the central representation of $L(N, 1)$ has $\mathrm{cs} = -N/4$. Modulo $\frac{1}{2}$ the formula agrees with SnapPy to 10^{-6} on 482 cone paths. The mod-1 refinement has not been checked numerically.

Theorem 5 (spin–Schläfli criterion)**.** *Let K be hyperbolic with $\mathrm{Arf}(K) = 0$. Suppose the cone paths from the complete structure to the ± 2 fillings exist, and set*

$$\Delta_K = \frac{1}{4\pi^2} \int_0^{2\pi} \big(\tau_2 - \tau_{-2}\big)\, d\alpha.$$

If $|\Delta_K| < \frac{1}{2}$, then $S^3_2(K) \ncong S^3_{-2}(K)$. It suffices that $\sup|\tau_2| + \sup|\tau_{-2}| < \pi$.

3. Evidence and limits

- *Survey of knots up to 10 crossings.* On the 217 knots whose cone paths are clean, $|\Delta| \leq 0.135$ (median 0.014) and $\sup|\tau_2| + \sup|\tau_{-2}| \leq 2.72 < \pi$.
- *Genus-2 knots with $\Delta_K = 1$, up to 13 crossings.* These are KT, $12n_{313}$, $12n_{430}$, $13n_{71}$, $13n_{1019}$, $13n_{1756}$, $13n_{1757}$ and $13n_{4591}$. For them $\Delta_K \in [0.028, 0.076]$. These checks are numerical and add nothing new, since Futer–Purcell–Schleimer already cover ≤ 19 crossings.
- *Heuristic.* At leading order in Neumann–Zagier, $\Delta_K \approx \frac{1}{2}\big[(2+x)/|\tau+2|^2 - (x-2)/|\tau-2|^2\big]$, where $\tau = x + iy$ is the cusp shape. This gives $|\Delta_K| \lesssim 1/(2\,\mathrm{Im}\,\tau)$, so a cosmetic pair would need a cusp with $\mathrm{Im}\,\tau \lesssim 1$ or a large twist.

- *What it is.* It reduces the conjecture, for these knots, to an *a priori twist bound*: the cone paths must not accumulate half a period of twist. The spin input is what is specific to S^3 and to ± 2. For $\pm 1/q$ there is only one spin structure and no gap. Ren has shown that purely cosmetic fillings do exist on some multi-cusped hyperbolic manifolds, so any proof needs such input.
- *What it is not.* It is not uniform. The ± 2 slopes are short (normalised length about 2.8 for KT), outside the range where Hodgson–Kerckhoff guarantee that cone paths exist, and no a priori twist bound is known.
- *Gauge-theoretic reading.* The useful specialisation is $\theta = 0$: the real part of complex Chern–Simons, taken for the spin-lifted hyperbolic holonomy. Volume ($\theta = \pi/2$) cannot see orientation.